A focused reproduction of the MHC Hammer exon-5 splicing axis in TCGA-BRCA HLA-A/B/C: an underpowered null and a characterized constitutive-splicing confounder

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Internal report. Sources: `METHODS.md`, `RESULTS_SUMMARY.md`, `PRESENTATION_NOTES.md`, `outputs/FINAL_REVIEW.md`, `OVERNIGHT_LOG.md`, `outputs/PHASE1_BC_strand_check.md`, `outputs/PHASE3_pancancer_descriptive.md`, `outputs/PHASE5_pancancer_matched_normal.md`, `outputs/FIGURE_AUDIT.md`, `ANNOTATION_ARTIFACT_NOTE.md`, `outputs/scratch/OPEN_QUESTIONS_DIAGNOSTIC.md`, and the CSVs under `outputs/`. Every number below traces to one of these files; where a number is not available from them, it is marked `[verify: ...]` rather than estimated.

Abstract

We built a schema-first, statistically explicit pipeline to test two specific alternative-splicing (AS) events at the HLA class I exon-5/intron-5 boundary — complete exon-5 skipping and partial intron-5 retention — reported by MHC Hammer (Puttick et al., *Nat Genet* 2024) as pervasive in normal tissue and, within-patient, enriched in a subset of tumors. Using TCGA-BRCA STAR splice-junction calls for HLA-A, HLA-B, and HLA-C (935 tumor / 106 normal samples) and a deliberately strict, high-confidence detection threshold, we find a rigorously power-characterized **null** result for both target events in both testable genes (HLA-A, HLA-C): no tumor-vs-normal difference survives Benjamini-Hochberg FDR correction anywhere in the project, and achieved statistical power for the two headline comparisons ranges from 9.8% to 29.4% — this is an honestly-quantified "cannot tell," not "no effect." Alongside the null, the project's strongest finding is methodological: a constitutive-annotation-artifact check caught HLA-A*11:01 carrying a near-universal (96.7–100% of carriers), tumor-independent minor splice isoform that would have looked like a promising cancer signal in a tumor-only design, and — on closer inspection prompted by this report — very likely a second such constitutive event in HLA-C (`C*03:04`/`C*03:03`, truncated by our detection threshold rather than genuinely rare). This is a reproduction that returns an honestly power-characterized null plus the identification of allele-specific constitutive splicing as a confounder for this kind of analysis — **not** a positive replication of the paper's tumor-enrichment finding, and it should not be read or presented as one.

Background

Biology. HLA class I genes (HLA-A/B/C) undergo alternative splicing at their exon-5/intron-5 boundary. Complete exon-5 skipping removes the transmembrane-anchoring exon, producing a soluble HLA isoform; partial intron-5 retention is a partial retention of intron 5 near the same boundary. MHC Hammer (Puttick et al., *Nat Genet* 2024) profiled HLA splicing across large normal and tumor cohorts and reported: (a) HLA splicing is pervasive in **normal** tissue, with exon-5

skipping the single most common event (76% of normal lung samples, 69% of normal breast samples show it); (b) the large majority of individual splicing events (~86–90%) occur at low abundance — below 10% of novel-transcript proportion; (c) HLA-C*03:04 shows an allele-specific, essentially constitutive (~100% of carriers, 70/70 lung and 57/57 breast) in-frame partial intron-5 retention in normal tissue; and (d) tumor enrichment of AS, when it occurs, is a **within-patient** phenomenon — compared against that same patient's matched adjacent-normal sample, present in an estimated 15% of ER+ breast tumors — and the paper states matched normals are required to make that call.

What this project set out to test. A focused reproduction of two of the paper's four reported AS classes — complete exon-5 skip (`exon5_skip`) and partial intron-5 retention (`intron5_retention`) — in TCGA-BRCA, extended to all three classical class I genes (HLA-A/B/C) and, where the data allow, to several pan-cancer cohorts. The paper's other two classes (partial exon skip, complete intron retention) and other splice axes it reports (e.g. exon-3 skipping) are **out of scope here**; complete intron retention was not observed at all in the paper's own data and is not tested here either.

Design difference, stated up front. The paper's tumor-enrichment claim is **within-patient**, tumor vs that same patient's matched adjacent-normal. This project's BRCA comparison is an **unpaired cohort proportion** — tumor samples vs a normal cohort that is not per-patient matched to the tumor cohort (935 tumor samples vs 106 normal samples, not 106 matched pairs). This is a materially different, weaker design for detecting a within-patient effect, and every tumor-vs-normal number in this report should be read with that in mind. This project does not attempt and does not claim a replication of the paper's within-patient result.

The contribution. Beyond re-running the paper's target events, this project's addition is a formal normal-tissue comparator with full statistical accounting (Wilson CIs, odds ratios, Fisher exact tests, BH-FDR across patterns, and an explicit power/minimum-detectable-effect analysis) plus a constitutive-annotation-artifact check that is not part of the paper's published method as reproduced here. That check is what surfaced this project's two most important findings (below).

Methods

Full detail, thresholds, and line-level code references: `METHODS.md`. Summary, in pipeline order:

1. **Upstream, outside this repo (not verified here).** Raw TCGA-BRCA BAMs sliced to the MHC region, personalized per-patient HLA-A/B/C reference contigs built from each patient's own genotype, STAR-aligned to produce one `SJ.out.tab`-style junction file per sample × allele, then exported to the flat 13-column TSVs this repo reads. **Two steps here are explicitly unconfirmed inferences, not verified facts**, and are labeled as such throughout this report:

    - *HLA typing tool.* The per-patient two-field HLA-A/B/C genotype was most plausibly generated by a tool such as arcasHLA, inferred only from the column shape of `data/gdc_sample_sheet_normal_with_hla.tsv` and common practice — **no typing-tool name, version, or invocation appears anywhere in this codebase.**

- *HLA-B/C reference orientation.* HLA-B/C are minus-strand genes, yet the exon5_skip/intron5_retention geometry (written in transcript-relative, effectively plus-strand terms) produces a coherent signal for HLA-C **without** a strand-flip transform (Methods §3d below). The working explanation — that personalized HLA-B/C reference contigs were likely already reverse-complemented at construction time — is an **inference**, never confirmed against the actual reference-construction code, which is not present in this repo.

2. **Per-sample derived columns** (`analyze_brca.compute_derived_columns()`): cumulative-sum bucketing of each row under its nearest upstream annotated junction (`Annotated_cumsum`), and `Local_rate` — a novel junction's share of `Uniquely_mapping` reads among every row competing in its own bucket.

3. **AS-event detection** (`find_as_events()`): a row qualifies as a candidate AS event if `Annotated == 0` (novel junction), `Annotated_cumsum != 0` (occurs downstream of a real annotated junction), and `Local_rate > 0.10` — the novel junction must carry more than 10% of its local bucket's reads. **This 10% floor is a deliberate, stricter-than-the-paper choice.** The paper finds ~86–90% of its splicing events occur *below* a 0.1 novel-transcript-proportion cutoff; this project's threshold sits exactly where the paper says most of the signal is *excluded*. The tradeoff is explicit: high confidence that anything detected is a real, non-noise event, at the cost of low sensitivity to the low-abundance majority of events the paper actually reports. This tradeoff is a throughline of the Results below — it has two demonstrated downstream consequences (Results §5 and §4).

4. **Geometry classification** (`classify_events()`): parses each event as `W1_W2 -> V1_V2` (wild-type → variant junction) and calls `exon5_skip` when the donor is unchanged and the acceptor jumps ≥100 bp downstream (`exon_skip_min_jump = 100`), or `intron5_retention` when the acceptor is unchanged and the donor shifts 1–50 bp upstream (`intron_retention_max_shift = 50`). Both are coordinate-offset judgment calls, not independently re-derived per gene against measured exon-5 length — no reference/GTF file with exon boundaries exists in this repo to do so (`outputs/scratch/OPEN_QUESTIONS_DIAGNOSTIC.md`).

5. **Strand-orientation validation (Phase 1).** Because HLA-B (~97.7–97.8% minus-strand) and HLA-C (~94.4–94.7% minus-strand) junctions might need a strand-aware coordinate mirror before the plus-strand-derived geometry above applies correctly, both genes were classified two ways — naive and strand-mirrored — and compared empirically. **HLA-C: naive classifier validated (`valid_as_is`)** — it produces a coherent, plausible signal (30 carriers vs 0 for the flipped version, which destroys nearly all of it). **HLA-B: neither scheme could be validated (`untrustworthy`)** — only 2 candidate AS events exist in the entire 935-tumor/106-normal cohort at this threshold (max observed `Local_rate` 12.6%, barely above the 10% floor, vs 52–60% for A/C), too few to discriminate the two schemes. Consequently **HLA-B is quarantined**: every HLA-B output is written under an `outputs/UNVALIDATED_HLA_B_*` prefix, excluded from every presented figure/table in this report, and is not un-quarantined here.

6. **Constitutive-annotation-artifact check.** For alleles with ≥20 tumor+normal carriers, find each allele's single most-recurrent AS pattern and its penetrance (% of that allele's own carriers showing it) in tumor and normal separately. An allele is flagged `constitutive_annotation_artifact` when penetrance is ≥80% in every group with usable data — i.e.,

essentially everyone who carries the allele shows the event, tumor or normal alike, which carries no tumor-vs-normal information regardless of statistical significance.

7. **Tumor-vs-normal statistics**, identical across every gene/cohort: Wilson 95% CIs on carrier proportions, risk difference and odds ratio (Haldane-Anscombe corrected when a 2×2 cell is 0) with CIs, Fisher exact tests, and Benjamini-Hochberg FDR across every tested per-pattern comparison (`min_carriers` $\geq$ `3`).

8. **Power analysis.** Two-proportion z-test achieved power (Cohen's h) for the observed effect, plus the minimum detectable effect at 80% power given the actual n=935/106 — reported as an honest "how underpowered," not a pass/fail label.

Every threshold used, and whether it was re-tuned per gene

Parameter	Value	Re-tuned for B/C or pan-cancer?
`local_rate_threshold`	0.10	No — identical everywhere
`exon_skip_min_jump`	100 bp	No — inherited from HLA-A
`intron_retention_max_shift`	50 bp	No — inherited from HLA-A
`min_allele_samples` (BRCA scale)	20	No
`min_allele_samples` (pan-cancer per-cohort)	10	Lowered deliberately — smaller cohorts
`min_carriers` (per-pattern Fisher/FDR)	3	No
`exclude_alleles` (HLA-A)	`hla_a_11_01`	Re-derived per gene for B/C; none flagged
Constitutive-artifact gates	`HIGH_PENETRANCE=80%`, `LOW_PENETRANCE=5%`, `min_n=10`	No

Full table with rationale: `METHODS.md` §9.

Results

1. The pipeline generalizes across genes and cohorts

The HLA-A pipeline, unmodified, was pointed at a second multi-cohort file (`data/pan_HLA-A_SJ_combined.tsv`, 9,738 rows — direct file count — across 6 cohorts, no BRCA) with zero code changes and correctly loaded, bucketed, and classified 46 candidate AS events (`outputs/PHASE3_pancancer_descriptive.md`). Distinct-sample count: 301 distinct barcodes by direct count of the file; `outputs/PHASE3_pancancer_descriptive.md`'s own summary states "311 total pan-cancer samples," which double-counts 10 barcodes that the file labels under more than one `Cohort` value — both numbers trace to the same underlying file, they are just counted two different ways, and the

discrepancy is noted here rather than silently resolved one way. This is a capability demonstration, not a biological claim; it established the pipeline was ready to be pointed at HLA-B/C and pan-cancer cohorts without gene-specific rewrites — **an engineering generalization only**: this check confirms the code runs on a new file, not that the exon5_skip/intron5_retention analysis is valid for every gene it can run on. Per-gene analysis validity was checked separately (Methods §5), and HLA-B failed that check (§6) even though the same code executed without error on HLA-B's data too. *Figures:* `figures_overview.pdf`; no dedicated figure for this check.

2. BRCA tumor-vs-normal: a null, led by its power, not by its p-value

Lead with the power, not the p-value. Both headline comparisons in BRCA are underpowered to detect an effect of the size the paper reports, and we can say precisely how underpowered (`outputs/table_power_analysis.csv`):

Event class	Tumor	Normal	Fisher p	OR (95% CI)	Achieved power	MDE at 80% power
exon5_skip (A*11:01 excluded)	0.1% (1/935)	0.0% (0/106)	1.0	0.34 (0.01–8.45)†	**9.8%**	tumor rate would need to fall outside [0.0%, 2.0%]
intron5_retention (A*11:01 excluded)	4.2% (39/935)	7.5% (8/106)	0.13	0.53 (0.24–1.17)	**29.4%**	tumor rate would need to fall outside [1.8%, 16.8%]

† Haldane-Anscombe corrected (a 2×2 cell was 0). Note: at 1 tumor carrier vs 0 normal carriers, exon5_skip is effectively uninterpretable as a direction either way — the raw proportions (0.1% > 0.0%) point "enriched in tumor" (as labeled on `outputs/fig_classes.pdf`), while the Haldane-corrected OR (0.34, <1) points "depleted." Both are correct arithmetic on the same 1-vs-0 cell counts; neither should be read as a real direction. See §5 for why this event class is near-absent at our threshold in the first place.

At 9.8% and 29.4% achieved power, this cohort's normal side (n=106) cannot reliably detect an effect the size the paper reports in **either** direction. `intron5_retention`'s point estimate trends *depleted* in tumor — the opposite of the paper's predicted enrichment — but with a 95% CI that crosses 1 (0.24–1.17) and 29.4% power, this is honestly "cannot tell," not a contradiction of the paper. A 12-setting one-at-a-time sweep of all three judgment-call thresholds keeps the direction "depleted in tumor" in all 12 settings (OR range 0.41–0.60), so the null is not an artifact of one specific cutoff choice. **0 of 6 tested per-pattern comparisons survive BH-FDR q<0.05** in the full per-pattern table. *Figures:* `outputs/fig_classes.pdf`, `outputs/fig_threshold_sensitivity.pdf`, `outputs/fig_volcano.pdf` (nominal-vs-FDR significance line labeled per `outputs/FIGURE_AUDIT.md`), `figures_overview.pdf` panels A1/C.

3. HLA-A*11:01: a constitutive splice isoform, not tumor biology — the strongest finding in the project

The `hla_a_11_01` allele's top recurrent event (`2365_2806->2383_2806`, classified `intron5_retention`) is present in **96.7% of tumor carriers (118/122)** and **100% of normal carriers**

(19/19) — no other allele in the cohort comes close, and the rate is identical whether the sample is tumor or normal (`ANNOTATION_ARTIFACT_NOTE.md`). A read-share check rules out the simplest explanation (a reference-coordinate swap): the **annotated** junction stays the majority isoform at ~72.4% of local reads on average in both groups, so this looks like a real, fixed, ~25–30%-penetrant minor splice isoform of A*11:01's reference transcript — a property of the allele, not of cancer status. In a tumor-only design (no matched normal comparator), this allele's high recurrence alone would have looked like a promising cancer-associated signal; having normals is what caught it. Consequently `hla_a_11_01` is excluded from the headline/per-allele comparisons throughout (never from descriptive outputs). A pan-cancer descriptive check (not a new significance test — no per-cancer-type matched normal for this comparison) found the same junction recurs at 100% (20/20) in the pooled normal-tissue reference, 100% (3/3) in TCGA-CHOL, and 85.7% (6/7) in TCGA-LGG (TCGA-HNSC 1/3 is a small-n outlier); pooled across all pan-cancer cohorts, 30/33 (91%) of A*11:01 carriers show the pattern — this is not a BRCA-specific quirk. *Figures:* `outputs/fig_per_allele.pdf` / `outputs/fig_per_allele_talk.pdf`, `figures_overview.pdf` panels B and D.

4. HLA-C: very likely a *second* constitutive artifact, truncated by our threshold — not tumor-enrichment [RESOLVED]

HLA-C's own recurrent candidate — `hla_c_03_04`/`hla_c_03_03`, pattern `1273_1711->1291_1711`, classified `intron5_retention` — was previously reported in this project's other documents (`RESULTS_SUMMARY.md` Finding 4/6) as a `real_variable_splicing_candidate` with a tumor-enriched odds ratio (OR 3.37) and cross-cohort directional corroboration. **That framing is retired here.** MHC Hammer reports HLA-C*03:04 partial intron-5 retention as an essentially constitutive event — present in 100% of carriers (70/70 lung, 57/57 breast) in normal tissue (Puttick et al., *Nat Genet* 2024). Our measured penetrance for the *same allele, same intron-5-retention geometry* is far below that: 19.33% tumor (23/119) / 11.11% normal (1/9) for `hla_c_03_04`, and 13.75% (11/80) / 7.14% (1/14) for `hla_c_03_03` (`outputs/scratch/OPEN_QUESTIONS_DIAGNOSTIC.md`, cross-checked against `outputs/table_per_allele_HLA-C.csv`). The gap between our ~11–19% penetrance and the paper's ~100% is exactly in the direction our 10% local-rate floor would produce if the true event is constitutive but frequently expressed below that floor (Methods §3, the same tradeoff already flagged there) — i.e., **very likely the same constitutive event Finding 3 already found for A*11:01, a second instance of the same detection-threshold confound, not real tumor-enriched splicing.**

We state this as "very likely," not "is." The one remaining confirmation — mapping coordinates 1291/1711 on the personalized HLA-C*03:04 contig to intron 5 to verify geometric identity with the paper's event — **could not be done**: no reference contig, GTF, or feature-table file for the personalized HLA-C*03:04 sequence exists anywhere in this repository (`outputs/scratch/OPEN_QUESTIONS_DIAGNOSTIC.md`). This is the single open follow-up for this finding (see Follow-ups).

Consequence for prior framing: the previously-reported OR of 3.37 for HLA-C `intron5_retention` (Results §2-style headline test, all C alleles pooled) rested on a single normal-side carrier (1/106) and was already non-significant (95% CI 0.45–25.0). Per-allele, the same pattern is non-significant for both alleles individually (Fisher p=0.69 for `hla_c_03_03`, p=1.0 for `hla_c_03_04`). None of this was ever a positive finding by this project's own statistical bar; it is now better understood as a **cautionary near-miss** — the constitutive-artifact check (Results §3/Methods §6) would have flagged this allele the same way it flagged A*11:01, had the 10%-threshold-

truncated penetrance not sat below the check's 80% gate. *Figures:* `outputs/fig_classes_HLA-C.pdf`, `outputs/fig_per_allele_HLA-C.pdf` / `outputs/fig_per_allele_talk_HLA-C.pdf`, `outputs/fig_volcano_HLA-C.pdf`.

5. exon5_skip near-absence: confirmed threshold-driven, not biological [RESOLVED]

HLA-A's `exon5_skip` is nearly unobserved at the presented 10% threshold (1 distinct tumor carrier, 0 normal). Because the paper reports exon-5 skipping as its *single most common* event genome-wide (76% normal lung, 69% normal breast) and ~86–90% of events occur below a 0.1 novel-transcript-proportion cutoff, this near-absence was an open question: biological, or an artifact of sitting at exactly the threshold the paper says excludes most signal?

An in-memory sensitivity diagnostic — the pipeline's own unmodified `load_data` → `compute_derived_columns` → `find_as_events` → `annotate_events` → `classify_events` functions, re-run with `local_rate_threshold=0.02` passed as an explicit argument (no config file, default, or output of record changed) — raised `exon5_skip` from **1 tumor / 0 normal distinct carriers at 0.10** (2 candidate rows collapsing to 1 distinct carrier, consistent with §2) to **44 tumor / 7 normal distinct carriers at 0.02** (`outputs/scratch/OPEN_QUESTIONS_DIAGNOSTIC.md`). The 0.10-threshold counts in this diagnostic match the presented pipeline's own output exactly, confirming the diagnostic used the identical code path, not a divergent reimplementation.

Conclusion: exon5_skip's near-absence at our 10% threshold is a threshold/sensitivity effect, consistent with the paper's own low-abundance regime for this event class — not biological absence. We present this as **confirmed by diagnostic**, not as a validated alternative analysis: no 0.02-threshold result set exists of record, no output under `outputs/` was regenerated at 0.02, and no figure/table in this report uses the 0.02 run. It answers "is the null real or a threshold artifact" without proposing 0.02 as a replacement threshold. The exon-5-length check that would have been an independent sanity check on this remains unavailable (no GTF/reference file in this repo) but is not needed given this result.

6. HLA-B: quarantined, insufficient signal at our threshold

HLA-B produces only 2 candidate AS events in the entire 935-tumor/106-normal BRCA cohort at the 10% local-rate threshold (vs 60 for HLA-C, ~200+ for HLA-A) — its maximum observed `Local_rate` in this cohort is 12.6%, barely above the detection floor, vs 52–60% for A/C. Neither the naive nor the strand-mirrored classifier could be validated against this little data (Methods §5). Whether this ceiling reflects real biology (less alternative splicing at HLA-B) or a technical/reference-quality issue specific to B is **not determined** by this project. All HLA-B output exists for audit under `outputs/UNVALIDATED_HLA_B_*` and is excluded from every result and figure above; it is not un-quarantined in this report and should not be cited as a finding, positive or negative.

7. Phase 5: a reusable matched-normal capability, not a solved power problem

A separate GDC sample sheet (740 normal samples, 24 TCGA projects, with HLA-A/B/C typing) let this project resolve the pan-cancer files' pooled, cohort-undifferentiated `NORMAL` block back to each sample's true TCGA project of origin (145/145 barcodes resolved for HLA-A, 318/318 for HLA-B/C) — turning a purely descriptive pan-cancer pass into genuine matched (same cancer type)

tumor-vs-normal tests, using the identical Fisher/Wilson-CI/odds-ratio machinery as the BRCA headline test, wherever a resolved cohort has ≥3 samples on both sides: TCGA-CHOL (33 tumor/8 normal) and TCGA-HNSC (52 tumor/5 normal) for HLA-A; TCGA-CHOL (33/8), TCGA-HNSC (67/35), and TCGA-KIRC (92/37) for HLA-C (HLA-B computed identically but quarantined per §6).

This should be read as a demonstrated *capability* — the pipeline can run a real matched-normal test on any cohort a typing sheet resolves — not as evidence the power problem is solved. The newly-matched normal sides are themselves small (as few as 5 samples, TCGA-HNSC/HLA-A) and, like BRCA, are unpaired cohort proportions, not within-patient pairs. All 10 presented (gene × cohort × class) tests come back null — every OR 95% CI spans 1, 0 have an OR CI excluding 1 at the nominal level (`outputs/table_phase5_summary.csv`, `outputs/FIGURE_AUDIT.md`) — consistent with, not independent confirmation beyond, the BRCA-scale power ceiling in §2. One methodological catch is worth noting: an early version of this pass re-derived the A*11:01 exclusion per cohort (requiring ≥10–20 carriers to flag anything) and missed it in TCGA-CHOL/TCGA-HNSC, where only 2–3 tumor samples carry the allele — 100% of the *original* HLA-A TCGA-CHOL/TCGA-HNSC "enrichment" signal turned out to be A*11:01's already-known artifact reappearing. Fixed by excluding A*11:01 unconditionally, regardless of per-cohort carrier count. *Figures:* `outputs/fig_phase5_matched_normal.pdf` (tested/significant count annotated per `outputs/FIGURE_AUDIT.md`).

Discussion

Three of the results above share one cause. The project's 10% local-rate threshold was chosen deliberately for high confidence (anything detected is almost certainly not alignment noise) at the acknowledged cost of low sensitivity — the paper itself reports ~86–90% of real events sit below that cutoff. That single choice explains **both** of this report's two resolved open questions: exon5_skip's near-total absence at 10% (§5), which a 0.02 diagnostic shows is a threshold effect, not biology; and HLA-C's `C*03:04/C*03:03` candidate's measured penetrance (11–19%) sitting far below the paper's reported ~100%, in exactly the direction a threshold-truncated constitutive event would produce (§4). Put together with the A*11:01 finding (§3), the throughline is: **this project's real methodological contribution is not a tumor-vs-normal effect size — it is a constitutive-annotation-artifact check, now demonstrated on two independent alleles across two genes, that would otherwise have been mistaken for biology in a design without matched normals.** A tumor-only version of this analysis would very plausibly have reported both A*11:01 and `C*03:04/C*03:03` as promising tumor-associated splicing hits; having normals, and building a formal penetrance-based artifact check rather than trusting recurrence alone, is what caught both.

None of this should be read as evidence against the paper's tumor-enrichment finding. The paper's claim is within-patient, matched-adjacent-normal, reported in an estimated 15% of ER+ breast tumors; this project's unpaired, cohort-level BRCA comparison — even before the threshold and constitutive- artifact issues above — was never powered to detect an effect of that shape (§2). The honest summary is: this reproduction neither confirms nor refutes the paper's tumor-enrichment claim, and identifies two mechanisms (threshold truncation, allele-specific constitutive splicing) that a future, better- powered, within-patient-paired attempt at this question would need to account for.

Limitations

- **Unpaired vs. within-patient design.** This project's BRCA (and Phase 5 pan-cancer) comparisons are unpaired cohort proportions; the paper's tumor-enrichment claim is within-patient, matched-adjacent-normal. No result here is a replication of that claim, positive or negative.
- **Detection-threshold tradeoff.** The 10% local-rate floor is stricter than the regime the paper reports most events occur in (~86–90% below 0.1). This is a deliberate high-confidence/low-sensitivity choice, and it is now demonstrated to materially suppress signal for at least two known events (exon5_skip generally, §5; HLA-C's `C*03:04/C*03:03` candidate, §4). No result-of-record in this project used a threshold other than 0.10.
- **Geometry thresholds not re-tuned per gene.** `exon_skip_min_jump` (100 bp) and `intron_retention_max_shift` (50 bp) were validated for HLA-A and carried over unchanged to HLA-B/C rather than independently re-derived against each gene's measured exon-5 length — no reference/GTF file with exon boundaries exists in this repo to do that re-derivation.
- **HLA-B is fully quarantined.** Insufficient signal at our threshold (2 candidate events total, max `Local_rate` 12.6%) to validate a classifier or test anything; real biology vs. a technical/reference-quality issue specific to HLA-B is undetermined. No HLA-B number in this report should be cited as a finding.
- **A known export bug was worked around, not fixed.** The pan-cancer HLA-B/C files contain a `Cohort == "BRCA_NORMAL"` block with `Type` mislabeled `"tumor"`; every downstream script excludes this block explicitly rather than relying on a corrected upstream export. Anyone reusing `data/pan_HLA-{B,C}_SJ_combined.tsv` without this project's notes would silently double-count BRCA normals as tumor samples.
- **Two upstream inferences remain unconfirmed.** The HLA typing tool (plausibly arcasHLA, inferred only from a sample-sheet column layout) and the explanation for why HLA-B/C's naive (unflipped) coordinate geometry behaves correctly despite both genes being majority minus-strand (the working theory: personalized reference contigs were already reverse-complemented at construction time) are both inferences about steps outside this repo, never confirmed against the actual upstream reference-construction or typing code.

Follow-ups

1. **Paired McNemar test on the 106 BRCA matched tumor/normal pairs.** The current headline test is an unpaired cohort proportion; a McNemar test on actually-paired samples would be a materially better-matched design against the paper's within-patient claim and does not require new data.
2. **Map HLA-C*03:04 coordinates 1291/1711 to confirm RESOLVED finding §4.** Requires the personalized HLA-C*03:04 reference contig and its exon/intron feature annotation from the upstream reference-construction step — not present in this repo; this is the single remaining confirmation for the "very likely a second constitutive artifact" call.

3. **Exon-3 skip and the paper's remaining AS classes** (partial exon skip; complete intron retention, not observed by the paper itself) — out of scope for this reproduction, open for a future pass.
4. **Confirm upstream reference construction and HLA typing directly** with whoever built the personalized per-patient HLA-A/B/C contigs: (a) whether HLA-B/C contigs were reverse-complemented at construction time (the inference this project's HLA-C classifier validation rests on), and (b) the actual HLA-typing tool/version used, before citing either upstream the way a publication would require.
5. **Resolve HLA-B's low event yield** — real biology vs. reference/alignment quality — as a prerequisite to un-quarantining any HLA-B result.